

Recommendation Systems Final Project – Innovation proposal

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1. Innovation Idea:

Our project aims to innovate on the recommender system outlined in the paper: *United We Stand, Divided We Fall*, by integrating a marketing-driven optimization strategy. We have selected the second suggestion from the innovation list. The focal point of this innovation is to prioritize and promote items with specific values in a chosen feature category, despite the dataset's anonymized feature categories. This approach could be useful, for example, to promote vendors paying a premium to boost sales of their items or to improve sales of a new item collection.

In this dataset, a subgroup is defined as items sharing specific feature values, denoted as A . We will denote the entire items set as X . Selecting A involves two assumptions: the subset is crafted randomly, without cherry-picking features and is large enough for statistically significant evaluations. We will conduct preliminary analysis to identify suitable feature category values based on prevalence.

According to the paper, the ensemble-based recommender system utilizes candidate selectors and rankers. We plan to adapt this algorithm to optimize the performance on group A , by potentially modifying these elements to favor marketing-aligned items. If candidate selectors already identify correct items, our focus will shift to refining rankers only. We propose enhancing ranker performance on subgroup A by adjusting the loss function, adding weights to incorrectly ranked item pairs where the lower-ranked item belongs to subgroup A .

2. Evaluation:

We will evaluate the retrained model's performance using the recommendation quality metric $MRR@100$. This metric gives a good indication as to how high the bought item is ranked. Our evaluation will compare the original performance of the specific subgroup before and after implementing the adapted algorithm. Additionally, we will assess the impact on the entire dataset's performance, as improvements in a specific subgroup might lead to decreased performance overall.

We may also want to consider comparing average recall, this metric could provide insight as to whether improving the candidate selectors is necessary. In this case the average recall will be calculated by taking all sessions with labels belonging to the subgroup, then checking if the label is included in the proposed candidates.

3. Experiments and Configurations:

We will perform an offline comparative analysis of performance on both the subgroup of items A , along with the entire items X prior and after retraining. The baseline model will include the UCF and GRU as the ensemble models for item candidates (best performing), with URM_W as the user rating matrix schema. Training data will contain the entirety of the 17 months, while the held-out test set for final performance evaluation will be the *test_final* set.